Cransfield Materials Testing Center
Quality Assurance

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PARTICLE SIZE DISTRIBUTION (BS 1377-2:1990: 9.2/9.3/9.4 & 9.5)

Client:	AHP					**Date submitted:**	2026-06-09
Project/Site Name:	KIRIAINI AHP					**Date tested:**	2026-06-11
Sample ID:	<0.063	**Sample No.:**	BH01	**Sample depth (M):**	0.0 - 25.0	**Sampled and Submitted by:**	Cransfield

Sample preparation details			**Moisture content at sample preparations**		
Initial quartered and air dried mass	(gm)	328	**Wet weight of sample**	(gm)	186.5
Washed and oven dried mass	(gm)	176.4	**Dry weight of sample**	(gm)	149.0
Fine mass	(gm)	151.6	**Wet weight of water**	(gm)	37.5
Fines percent	(%)	46.2	**Moisture content**	(%)	25.2

Soil Classification									
GRAVEL (%):	22.8	**SAND (%):**	5.3	**CLAY/SILT (%):**	71.9	**USCS**	MH - Silt of High Plasticity	**AASHTO**	A-7-5 (GI 19)

Wet & Dry Sieve Analysis to BS 1377-2:1990: 9.2/9.3/9.4

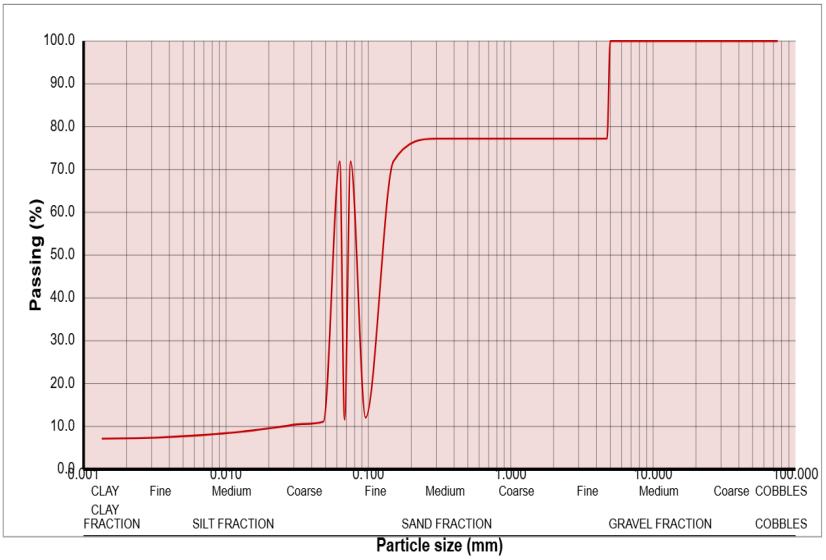
Sieve size (mm)	Retained mass (gm)	% Retained (%)	Cumulative passing (%)
75	0	0.0	100.0
63	0	0.0	100.0
50	0	0.0	100.0
37.5	0	0.0	100.0
28	0	0.0	100.0
20	0	0.0	100.0
14	0	0.0	100.0
10	0	0.0	100.0
6.3	0	0.0	100.0
5	0	0.0	100.0
4.75 (No. 4)	40.22	22.8	77.2
3.35	0	0.0	77.2
2.36	0	0.0	77.2
2.00 (No. 10)	0	0.0	77.2
1.18	0	0.0	77.2
0.6	0	0.0	77.2
0.425	0	0.0	77.2
0.3	0	0.0	77.2
0.15	9.35	5.3	71.9
0.075	0	0.0	71.9
0.063	0	0.0	71.9
<0.063	126.83	71.9	_____
TOTAL	**176.4**	**100.0**	

Hydrometer Analysis to BS 1377-2:1990:9.5

Dry weight (gm)	176.4	**S.G (Mg/m³)**	2.65
Hydro. type	152H	**Susp. vol. (cm³)**	1000
Zero corr. (g/L)	0	**Meniscus (g/L)**	0.1
Temp. (°C)	20	**K factor**	1.358

Time (min)	Actual HR (R n 2)	Adjusted (Rh)	Composite corr.	Corrected HR	Eff. depth (cm)	Diameter (mm)	% Fines (susp.)	% Fines (sample)
0.25	24.5	24.5	0.10	24.6	12.26	0.0956	22.4	12.0
0.50	_____	23.5	0.10	23.6	12.42	0.0680	21.5	11.6
1.00	22.6	22.6	0.10	22.7	12.57	0.0484	20.7	11.1
2.00	21.5	21.5	0.10	21.6	12.75	0.0345	19.7	10.6
4.00	20.2	20.2	0.10	20.3	12.97	0.0246	18.5	9.9
8.00	18.9	18.9	0.10	19.0	13.18	0.0175	17.3	9.3
15.00	17.8	17.8	0.10	17.9	13.36	0.0129	16.3	8.8
30.00	16.9	16.9	0.10	17.0	13.51	0.0092	15.5	8.3
60.00	16.1	16.1	0.10	16.2	13.64	0.0065	14.7	7.9
120.00	15.5	15.5	0.10	15.6	13.74	0.0046	14.2	7.6
240.00	15.0	15.0	0.10	15.1	13.82	0.0033	13.7	7.4
480.00	14.7	14.7	0.10	14.8	13.87	0.0023	13.5	7.2
1440.00	14.5	14.5	0.10	14.6	13.90	0.0013	13.3	7.1

PARTICLE SIZE DISTRIBUTION GRAPH



Tested by JILLO **Date reported** _____ **Checked by:** _____

CRANSFIELD MATERIAL TESTING P.O. BOX 59401-00100 NAIROBI